Artem Skoropadskyi

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PROFESSIONAL SUMMARY

C++ engineer with a real-time performance engineering background: concurrency, CPU-bound optimization, and memory-layout work under hard frame budgets in production game engine systems. Moved AI stat calculations off the game thread using Unreal Engine 5's task graph, keeping the render thread clear under load. Flattened class hierarchies and redesigned allocation patterns to cut 3D render load times and memory pressure. Built the interface bridge between a Python/PyTorch inference layer and a C++ engine runtime, reasoning about two systems with different clocks. Shipped production C++ on a **20+ person team** with 100+ commits, resolved 150+ engine-level bugs, and ran a formal QA sign-off process across 2 releases with zero critical post-release defects. Familiar with Linux via CS coursework and active toolchain (GCC/Clang/gdb) ramp-up; comfortable picking up new languages and platforms fast. BSc Computer Science. Maintains quantitative-reasoning practice through Jane Street Puzzle solving and competitive programming (LeetCode, top 600k global). English C1; open to relocating.

CORE COMPETENCIES

C++ (17/20/23) | Concurrency & Multithreading | Algorithms & Data Structures | CPU Architecture / Cache-Aware Design | Python | Performance Profiling | Memory Management | Reliability & QA Process

WORK EXPERIENCE

Donorom

Jun 2024 - Present

C++ Systems Developer (Freelance) · Poland, Remote

- Moved AI stat calculations off the game thread using UE5's task graph, keeping **7 gameplay features** thread-safe with zero data-race regressions under heavy tick load.
- Profiled and optimized a custom shading pipeline, cutting **GPU time by 17ms per frame** and improving average FPS from 23 to 60 in computation-heavy scenes.
- Architected a **data-driven JSON asset pipeline** processing 500+ runtime assets dynamically, eliminating hardcoded data dependencies.
- Built a JRPG combat system combining tactical positioning with cinematic sequences - authored all core C++ logic and Blueprint integration.

Luxolis.ai

Aug 2023 - Nov 2023

Software Developer - 3D / Rendering (Freelance) · Poland, Remote

- Flattened deep inheritance hierarchies and redesigned allocation strategy to cut **3D render load times and memory pressure** - replaced dynamic dispatch with static dispatch in hot paths.
- Improved rendering throughput via **custom shader optimization and strategic light baking**, achieving consistent frame-time reductions.
- Resolved **40+ cross-system integration bugs** between a C++ rendering layer and an AI/ML pipeline developed by a separate team, without owning the ML codebase.

Dragon's Lake

Mar 2022 - Feb 2023

C++ / Unreal Engine Developer · Ukraine, Remote

- Shipped production C++ systems on a **20+ person team**, managing 100+ commits across movement, AI, and combat subsystems.
- Resolved **150+ gameplay and engine-level bugs**, measurably improving project stability across 6 delivery sprints.
- Delivered **consistent frame rate improvements on mid-range hardware** through targeted rendering and AI logic optimizations.

PROJECTS

UE5 Task Graph - Thread-Safe AI Under Load

C++ · Concurrency

Moved AI stat calculation off the game thread using Unreal Engine 5's task graph API, using synchronization primitives to keep the render thread clear and avoid data races during heavy AI tick computation.

Tech: C++, UE5 Task Graph, async tasks, synchronization primitives

AI Model Integration - Real-Time API Bridge

Python · PyTorch · C++

Designed and built the interface bridge between a Python/PyTorch inference layer and a C++ engine runtime, managing real-time data flow across two systems with different clocks (engine tick vs. inference latency).

Tech: Python, PyTorch, StableBaselines3, Pandas (movement dataset prep), C++, Unreal Engine 5 · Repo: available on request

3D Rendering Optimization Toolkit

C++ · Memory / CPU

Performance overhaul of a 3D web visualization service. Flattened class hierarchies and redesigned memory allocation strategy, achieving consistent FPS improvements and reduced load times through cache-aware data layout changes.

Tech: C++, OpenGL, DirectX, custom HLSL shaders, performance profiling

Horizons of Achaea

Unreal Engine 5 · C++ · Steam

Shipped demo on Steam. Primary C++ developer for core systems: tactical battle engine, branching dialogue across 5+ sequences, JSON-driven asset pipeline (500+ runtime assets), custom shading pipeline. GPU frame time cut 17ms via shader profiling and AI logic rearchitecture.

Tech: Unreal Engine 5, C++, C#, Blueprints, Python, custom JSON library

EDUCATION

Bachelor of Science, Computer Science · NTU of Ukraine "Kyiv Polytechnic Institute" Sep 2020 - Jun 2024

Kyiv, Ukraine

SKILLS

Programming	C++ (primary, modern C++17/20), Python, C#, Blueprint visual scripting
Systems	Concurrency & multithreading (UE5 task graph, render/game thread separation), memory management, cache-aware data layout, CPU-bound profiling
CS Fundamentals	Algorithms and data structures, OS fundamentals (stack/heap, scheduling), constraint-based algorithm design (procedural generation)
Competitive Programming	LeetCode Global Rank ~600,000 (~top 2% of registered users)
Tools	Git, JIRA, GitHub, Unreal Engine 5/4 (C++, Nanite, Lumen), Linux (CS coursework, GCC/Clang/gdb toolchain familiarization)
ML / Research	PyTorch, StableBaselines3, TensorFlow; reinforcement learning; Pandas (dataset prep, diploma project)
Languages	Ukrainian (native), Russian (bilingual), English (C1), German (B2), Polish (B1)